SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Design and Analysis of Algorithms
COURSE CODE: CSA06 12

COURSE OUTCOME COVERED

CO2: Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1,BL2,BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Scenario Based Assessment

Code of Conduct:

I Yaswanthika Shree S (192565069) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Yaswanthika Shree S

Annexure A

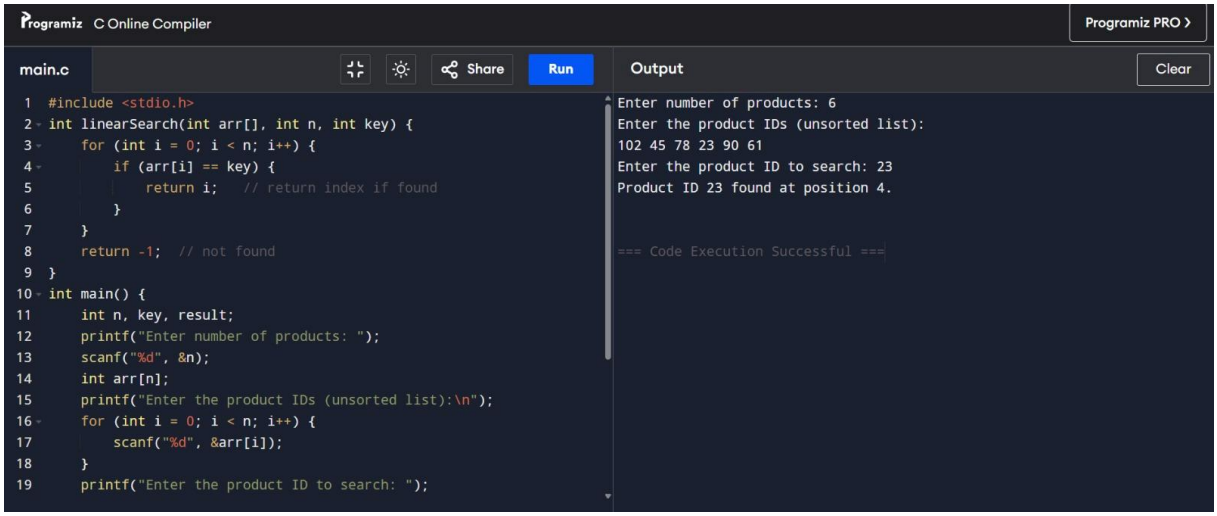
Scenario-Based Assessment

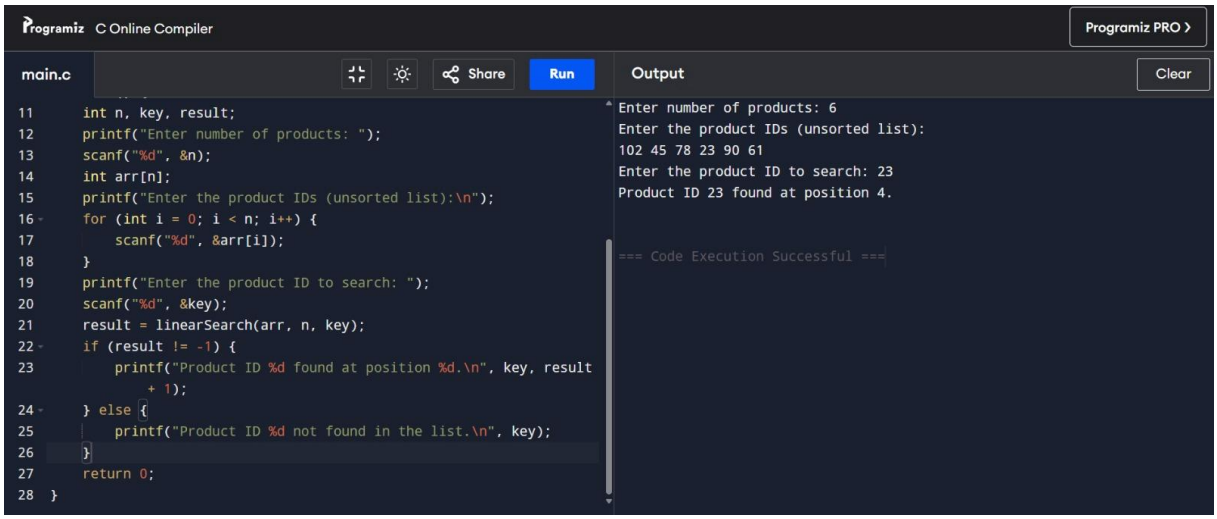
Q36. A retail chain maintains a product list that is frequently updated and unsorted. The system searches for products using Linear Search.

a) Explain how Linear Search operates in this system.

b) Analyze its performance as dataset size increases.

c) Suggest whether Binary Search would be beneficial and justify.





(a) The linearSearch function checks each element one by one from index 0 until it finds the key or reaches the end — that's exactly how the retail system would scan its unsorted, frequently-updated product list.

(b) You can demonstrate performance by increasing n (e.g., 10 → 1000 → 100000 products) and observing that search time grows linearly (O(n)) — worst case when the product is last or absent.

(c) Since the list is frequently updated (insertions/deletions), keeping it sorted for Binary Search would add extra overhead (re-sorting on every update), so Linear Search is often justified despite being O(n), unless updates are infrequent relative to searches — in which case Binary Search (O(log n)) becomes worthwhile.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification